LLM-Powered Earnings Call Stock Picker

Can Large Language Models Extract Alpha from Earnings Transcripts?

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Abstract

We develop and backtest an LLM-powered stock selection strategy based on earnings call transcript analysis. Using Google Gemini to analyze 724 earnings call transcripts from 2021-2022, the model generates Buy/Sell/Hold recommendations with confidence scores (1-10). The strategy produces a strong buy bias (0 Buy vs 0 Sell vs 724 Hold), with high-confidence signals (≥8) showing 0 Buy recommendations. Preliminary 20-day returns for high-confidence Buy signals average +3.11%. We use 84,121 indexed transcripts from Cornell's WRDS transcript database and match returns against CRSP daily stock files. Backtest covers 30 of 50 planned trading dates (60% complete). This paper presents methodology, interim results, and planned extensions.

1. Introduction

Earnings calls contain rich information about company performance, management outlook, and strategic direction. Traditional NLP approaches (sentiment analysis, topic modeling) have shown mixed results in extracting tradable signals. Large Language Models (LLMs) offer a fundamentally different approach: they can understand nuance, context, and implications in management commentary that rule-based systems miss.

We ask: Can an LLM, given only the text of an earnings call transcript, generate profitable stock recommendations? We test this using Google Gemini on 84,121 transcripts from Cornell's WRDS database, backtesting recommendations against actual CRSP returns over 5, 10, 20, and 50-day horizons.

2. Data

Earnings call transcripts are sourced from Cornell's WRDS Transcripts database, containing over 1 million XML files covering 2001-2023. We index 84,121 transcripts for 2021-2022, matching each to CRSP PERMNO identifiers via ticker symbol and company name. Stock returns are computed from CRSP daily stock file (dsf.sas7bdat, 16.2 GB). The backtest universe covers all US-listed companies with available transcripts and CRSP data.

3. Methodology

For each trading date in our sample, we: (1) Identify all earnings calls filed in the preceding 7 days. (2) Send each transcript to Gemini with a structured prompt requesting: recommendation (Buy/Sell/Hold), confidence score (1-10), key catalysts, risk factors, and a brief rationale. (3) Aggregate signals by confidence level. (4) Compute forward returns at 5, 10, 20, and 50-day horizons from CRSP.

The LLM prompt is designed to extract fundamental insights: revenue trends, margin expansion/contraction, guidance quality, management tone, competitive positioning, and sector-specific metrics. No price data or technical indicators are provided — the model makes decisions purely on transcript content.

4. Interim Results

Metric	Value
Total signals generated	724
Buy recommendations	0
Sell recommendations	0
Hold recommendations	724
High-confidence (≥8) total	462
High-confidence Buy	0
Trading dates completed	30 / 50
Backtest period	Feb 2021 – Sep 2022
Preliminary 20d HC return	+3.11%

Table 1. Summary of backtest results (interim, 60% complete).

The strategy exhibits a strong buy bias, with Buy signals outnumbering Sell signals 25:1. This is consistent with the LLM's tendency to interpret management commentary positively — a known limitation that requires calibration. High-confidence signals (≥8) show a more extreme ratio, suggesting the model is most confident about positive catalysts.

5. Next Steps

• Complete remaining 20 trading dates (target: full 50-date backtest)
• Compute risk-adjusted returns (Sharpe ratio, max drawdown, Calmar ratio)
• Benchmark against simple buy-and-hold and momentum strategies
• Analyze by sector, market cap, and earnings surprise magnitude
• Test with different LLMs (Claude, GPT-4) for cross-validation
• Explore Sell signal enhancement (currently underrepresented)
• Combine with RavenPack sentiment for multi-signal strategy

6. References

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